Literature Review — Music Works Catalogue Portal

Computer Science Final Year Project | 2025–2026

1. Introduction

This review examines seven bodies of work relevant to the Music Works Catalogue Portal — a project that replaces the broken MerMEId publication infrastructure with a modern, catalogue-agnostic web portal built on a REST API and an interactive interface. Sources span the foundational academic literature on thematic music cataloguing, the MerMEId system's rise and decline, broader digital humanities research on software sustainability, related platforms in digital musicology, and the technical framework selected for implementation. Where sources agree, this is noted; where they differ or reveal limitations, these tensions are discussed explicitly and used to motivate specific design decisions.

2. Thematic Catalogues and the MEI Standard

2.1 Krabbe & Geertinger (2012) — MEI as a Basis for Thematic Catalogues

Krabbe, N. & Geertinger, A.T. (2012). 'MEI (Music Encoding Initiative) as a Basis for Thematic Catalogues: Thoughts, Experiences, and Preliminary Results.' *RISM Conference on Music Documentation in Libraries, Scholarship, and Practice,* June 2012. Danish Centre for Music Publication.

The foundational document for this project, this paper describes the origins of MerMEId at the Danish Centre for Music Publication (DCM). Krabbe identifies the core failure of traditional printed thematic catalogues: they are frozen at the moment of publication, full of 'esoteric abbreviations and hundreds of library sigla', useful only to specialists. He argues convincingly that digital catalogues must go beyond PDF replicas of print editions to enable dynamic querying — concordances, chronological sorting, filtering by instrumentation, and other facets that static publications cannot support. Geertinger's technical contribution establishes MEI XML as the appropriate encoding standard: its metadata section can represent works, sources, manuscripts, incipits, performance history, and editorial commentary in a single open, community-maintained file. This argument has since become the field consensus — all subsequent work reviewed here, including Stadler et al. (2023) and the TROMPA project (Weigl et al., 2019), treats MEI as the accepted standard for scholarly music data encoding.

However, MEI's flexibility is also a structural liability. The standard's optional elements mean that different MerMEId projects encode equivalent information inconsistently, making cross-catalogue interoperability far harder than the designers intended. Krabbe and Geertinger acknowledge this only obliquely; Stadler et al. (2023) are more explicit about the consequences a decade later. This tension motivates the MEI evaluation deliverable in this project: rather than accepting MEI's catalogue module uncritically, this project will assess where it succeeds, where it creates unnecessary inconsistency, and what a revised data model might look like. Whether MEI actually delivered the interoperability it promised is, in practice, still an open question.

3. The MerMEId System — Achievement and Decline

3.1 Geertinger & Lundberg (2013) — The MerMEId Architecture

Geertinger, A.T. & Lundberg, S. (2013). 'MerMEId: Creating Thematic Catalogues Using MEI Metadata.' Poster, *Music Encoding Conference 2013,* Mainz.

This poster documents MerMEId as deployed in 2013: a web tool using XForms/Orbeon for editing and eXist-db as the XML database backend, recording source descriptions, performance histories, bibliographies, incipits, instrumentation, and more, across five catalogues at DCM. For this project it serves two purposes. First, it provides an accurate account of the MEI data model that the transformation pipeline must process — the structure of MerMEId XML files can be inferred from the system's documented capabilities. Second, it reveals the architectural choices that became liabilities:

the dependencies on XForms/Orbeon (a browser technology now superseded), eXist-db (a specialist XML database with a small developer community), and Java throughout were defensible in 2013 but have since made the system very difficult to maintain. Crucially, Geertinger and Lundberg do not discuss long-term sustainability — in 2013, it was not a visible concern. That absence is itself a lesson: this project's chosen stack (Python, SQLite, FastAPI) prioritises simplicity and mainstream adoption precisely to avoid the same institutional lock-in.

3.2 Bue, Gammert, Gubsch et al. / Stadler (2023) — Transitional MerMEId

Bue, M., Gammert, J., Gubsch, C., Jettka, D. et al. (2023). 'Transitional MerMEId – Responding to Community Needs in Software and Sustainability.' *TEI/MEC Conference,* Paderborn.

The most important single source for this project. The paper confirms that MerMEId is the only tool in digital musicology developed explicitly for managing MEI catalogue data, that significant community demand remains, and yet the codebase needs extensive revision: no comprehensive test suite exists, documentation is inadequate for both users and developers, and the software is not modularised. Critically, the community's response focuses on revising the authoring tool, not on building a new publication and access layer for end users. This creates the exact gap this project fills.

> *'MerMEId is an established — and the only — tool in digital musicology that has been developed explicitly for creating and managing musical catalogue data in MEI. [...] The code basis of the software needs extensive revision for re-usability and sustainability.' (Bue et al., 2023)*

A limitation of this source is that, as a conference abstract, it describes planned rather than completed revisions. This project therefore treats the underlying MEI XML data — not MerMEId itself — as its stable input, insulating the pipeline from ongoing tool instability.

4. Software Sustainability in Digital Humanities

4.1 Mathiak et al. (2020) — Sustainability Strategies for DH Systems

Mathiak, B., Neuefeind, C., Schildkamp, P. et al. (2020). 'Sustainability Strategies for Digital Humanities Systems.' *Digital Humanities Conference (DH2020),* Alliance of Digital Humanities Organizations.

This panel paper — the only source reviewed here from outside the MEI community — provides quantitative evidence that places MerMEId's collapse in a documented systemic pattern rather than treating it as an isolated failure. Mathiak et al. examined 466 digital scholarly editions and found that 376 had disappeared from the web, giving an average project lifetime of only 8.5 years. The causes — diminishing funding, institutional abandonment, and personnel turnover — mirror MerMEId's trajectory precisely. The authors call this the 'Digital Dark Age' and identify the most effective mitigation strategies as those that use simple mainstream technology stacks, adopt open standards, and avoid vendor or single-institution lock-in. All three principles directly inform this project's architectural choices. The authors also note that Docker containerisation is among the more promising strategies for sustaining research software — a recommendation incorporated into this project's deployment plan.

A limitation of Mathiak et al. is that their study examines text-based digital editions rather than music-specific systems. The particular constraints of music metadata (the complexity of the MEI schema, the specialist community) mean some conclusions may generalise differently to the musicology domain. Nevertheless, the structural drivers of failure they identify — single-institution dependency, obsolete technology stacks, absence of testing — are directly mirrored in MerMEId's documented problems (Stadler et al., 2023), validating the comparison.

5. Related Platforms in Digital Musicology

5.1 IMSLP — International Music Score Library Project

IMSLP — Petrucci Music Library (2006–present). [Online]. Available at: https://imslp.org [Accessed: 2025]. *Credibility: non-profit organisation; widely cited in peer-reviewed digital musicology literature, including Weigl et al. (2019).*

The world's largest public domain music score repository (256,000+ works, 27,600+ composers), IMSLP represents the most widely-used existing platform for accessing classical music online and is the natural comparator for this project from a general-user perspective. Its metadata model is deliberately shallow: works are listed by title and composer with PDF links, with no manuscript descriptions, premiere information, incipit encoding, movement-level detail, or scholarly commentary. This is not a criticism of IMSLP's purpose — score access for performers — but an observation about complementarity: where IMSLP provides breadth of access, this project provides scholarly depth. The two systems are designed to link rather than compete; this project's portal records will reference IMSLP entries via work identifiers where they exist, allowing users to move from catalogue metadata to score access in a single step.

5.2 MusicBrainz — Open Music Encyclopedia

MusicBrainz (MetaBrainz Foundation, 2000–present). [Online]. Available at: https://musicbrainz.org [Accessed: 2025]. *Credibility: registered non-profit; CC0-licensed open data; REST API documented at musicbrainz.org/doc/MusicBrainz_API; cited extensively in peer-reviewed literature.*

MusicBrainz is a community-maintained open music encyclopedia with a REST API, unique persistent identifiers (MBIDs), and Creative Commons-licensed data — the most technically accessible music database available for integration. Weigl et al. (2019) describe it as 'the main resource for public domain structured (and machine-readable) music metadata', noting its suitability for classical works including catalogue numbers and composer-work relationships. However, MusicBrainz's classical metadata is thin at the catalogue level: it records works and recordings but not manuscript sources, composition history, premiere dates, or incipit notation. Furthermore, community editing without philological review means data quality is variable. This project links portal records to MusicBrainz identifiers where they exist, enabling interoperability with the wider music data ecosystem, while providing the deeper catalogue-level data that MusicBrainz cannot hold given its community-editing model.

5.3 Weigl et al. (2019) — TROMPA and Music Data Interoperability

Weigl, D.M., Goebl, W., Crawford, T. et al. (2019). 'Interweaving and Enriching Digital Music Collections for Scholarship, Performance, and Enjoyment.' *6th International Conference on Digital Libraries for Musicology (DLfM),* The Hague, pp. 84–88.

The EU-funded TROMPA project describes an infrastructure for interconnecting disparate digital music collections using Linked Data and the FAIR principles. Its key insight is that individual repositories describe different facets of the same musical reality — a score on IMSLP, a recording on MusicBrainz, and a catalogue entry in MerMEId cannot currently be connected without manual effort — and that RDF and SPARQL provide the technically correct solution. TROMPA represents the state of the art for large-scale music data integration and its conclusions are well-supported. However, the full Linked Data stack requires triple stores, SPARQL endpoints, and ontology engineering that is beyond the scope of a final year project. This project incorporates TROMPA's core insight pragmatically: MusicBrainz and Wikidata identifiers appear in all API responses as forward-compatible linked data references, enabling future RDF migration without requiring it now. This is a deliberate, literature-informed design decision.

6. Technical Approach

6.1 FastAPI — Python REST API Framework

Ramírez, S. (2018–present). FastAPI. https://fastapi.tiangolo.com. Documented in: Lubanovic, B. (2023). *FastAPI: Modern Python Web Development.* O'Reilly Media. *Credibility: MIT licence; adopted by Netflix, Uber, Microsoft; O'Reilly technical publication (2023).*

FastAPI is chosen for the REST API layer. Its credibility is established via O'Reilly documentation and production adoption at major organisations rather than its own homepage. Compared to Flask — the most widely-taught Python API framework — FastAPI's automatic OpenAPI documentation generation from Python type hints is a decisive advantage: a well-documented public API is a core deliverable and directly addresses the absence of documentation identified by Stadler et al. (2023). Pydantic-based validation reduces the risk of serving malformed catalogue data; asynchronous architecture handles concurrent queries without blocking. The trade-off is relative youth (released 2018, smaller ecosystem than Django), which is not a significant constraint for a read-heavy JSON API over a SQLite backend.

7. Comparative Analysis

Source / Project	Type	Catalogue data	Public API	Scholarly depth
MerMEId / CNW (Krabbe & Geertinger, 2012; Geertinger & Lundberg, 2013)	Academic tool	✓ **Yes (MEI XML)**	✗ Broken / none	✓ **Full thematic catalogue**
IMSLP (imslp.org)	Open platform	✗ No thematic data	⚠️ Limited	✗ Score access only
MusicBrainz (musicbrainz.org)	Open encyclopedia	⚠️ Partial	✓ **REST/XML**	✗ Thin on classical detail
TROMPA / Weigl et al. (2019)	EU research project	⚠️ Linked data only	✓ **SPARQL**	⚠️ Broad, not deep
This project	Final year project	✓ **JSON/SQLite**	✓ **REST/JSON**	✓ **MEI catalogue depth**

Table 1: Comparison of related systems across key dimensions (sources as cited).

The table reveals a consistent pattern: no existing system combines structured thematic catalogue data, a modern public API, and the full scholarly depth of MEI-encoded catalogues. MerMEId has the data and depth but no working public interface (Stadler et al., 2023). IMSLP has broad reach but shallow metadata. MusicBrainz has a working API but lacks catalogue-level detail. TROMPA demonstrates what a fully linked-data solution looks like but at a scale and complexity beyond this project's scope. Mathiak et al. (2020) explain why MerMEId's failure was structurally predictable. This project occupies the unfilled position — the gap between available scholarly data and accessible modern interface — using a technology stack explicitly chosen to avoid the sustainability failures that the literature has documented.

8. Synthesis and Open Questions

The literature reviewed tells a coherent but cautionary story: the tools that best serve scholarly depth (MerMEId, MEI XML) have proven hardest to sustain and access, while the most accessible tools (IMSLP, MusicBrainz) lack depth. Mathiak et al. (2020) establish that this failure is not specific to musicology — it is systemic across digital humanities. Krabbe and Geertinger (2012) were right about MEI's potential but did not anticipate how institutional dependency would eventually break MerMEId. Stadler et al. (2023) document the consequences. Three questions remain unanswered by the literature that this project will address empirically: how much of the MEI catalogue data is actually parseable given documented encoding inconsistency; whether users find this type of portal genuinely useful (no user studies of thematic catalogue interfaces appear in the literature); and whether melodic similarity from incipits produces meaningful groupings or noise. These open questions represent this project's scholarly contribution — not merely implementing a known solution, but beginning to answer questions the field has not yet addressed.

9. References

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Lubanovic, B. (2023). FastAPI: Modern Python Web Development. Sebastopol, CA: O'Reilly Media. ISBN 978-1-098-13550-8.

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